

Ömer Ekmen

Software Engineering & Data Engineering

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Work authorization: Turkish national – eligible for EU Blue Card and Netherlands Highly Skilled Migrant sponsorship.

Profile

Engineer with 2.5 years across enterprise backend and the Microsoft data platform. Currently building production pipelines on Microsoft Fabric and Azure Data Factory at Koç Group; previously owned ERP accounting and purchasing modules in .NET, including a Framework to .NET 8 migration. B.Sc. Mathematics from Galatasaray University, taught in French, with a reinforcement learning thesis graded 4.0/4.0.

Technical Skills

Languages: C#, Python, SQL (T-SQL)

.NET: ASP.NET Core Web API, .NET 8, Entity Framework Core, LINQ, MediatR/CQRS, .NET Framework

Python: Django, FastAPI, SQLAlchemy, Pydantic, Poetry, Pandas, NumPy

Data: Microsoft Fabric (Lakehouse, Warehouse), Azure Data Factory, Azure SQL Database, Azure Functions, Azure Monitor, SSIS, SSRS, Power BI, MSSQL, PostgreSQL

Platform: Docker, Kubernetes, RabbitMQ, Redis, Azure DevOps, GitHub Actions, AWS (EC2, S3), Git, CI/CD, Linux

Project experience: Java 21, Spring Boot, Spring Cloud, Spring Data JPA, Spring Security, Maven, Flyway, Kafka, OpenTelemetry, PyTorch, scikit-learn, LangChain/LangGraph, vector databases (pgvector, Qdrant, Pinecone), Hugging Face, MCP

Working knowledge: Spark/PySpark, Parquet, Delta tables, dbt, Airflow, BigQuery, Polars, Celery

Experience

Data Science Engineer

Jul 2026 – Present

SeturTech (Koç Group) via Rise Technology, Istanbul

- Build and operate production data pipelines on Microsoft Fabric (lakehouse and data warehouse) and Azure Data Factory for Setur, Koç Group's travel and tourism company.
- Integrate source systems including Dynamics 365, SAP and CRM platforms into governed analytical models.
- Deliver specialised reporting for every business unit – C-level, finance and accounting, customer support and their leads – via Power BI, Azure SQL Database, Python notebooks and Azure Functions.
- Replace recurring manual extracts with scheduled, automated pipeline orchestration.

Software Engineer – Working Student (24 months)

Jun 2024 – May 2026

Otokoç Otomotiv (Koç Group), Istanbul

- Migrated production ERP modules from .NET Framework to .NET 8, maintaining parallel Framework support so operations continued uninterrupted.
- Developed accounting and purchasing backend modules using ASP.NET Core Web API, Entity Framework Core, LINQ and MediatR/CQRS.
- Automated personnel expense processing (business travel, lodging) via rule-based integration between Otokoç and Setur systems, reaching 70%+ time reduction; approximately 30% process time reduction across accounting and purchasing overall.
- Authored 50+ T-SQL stored procedures; built 6 SSIS ETL packages and 15+ SSRS reports.
- Tuned query performance via execution-plan analysis, indexing strategy and schema-level redesign on Microsoft SQL Server.
- Integrated RabbitMQ messaging and Redis caching; containerised with Docker and managed Kubernetes deployments alongside automated pipelines.
- Progressed from a team of 3–4 to sole engineer on the module for two months after teammates resigned, sustaining delivery without handover gaps.
- Shipped 20+ production-tracked items in Agile/Scrum via Azure DevOps, translating business requirements documents into technical tasks.

Freelance Software Engineer

Dec 2023 – Feb 2025

Independent, Remote

- Delivered two production systems for paying clients, owning requirements analysis, backend architecture,

implementation, CI/CD and AWS deployment.

- **Property management platform** (Nesin Foundation, Turkish mathematics education non-profit; 2-person team): Python/Django backend managing 100+ property listings with rent, expense and tenant workflows; PostgreSQL schema of 15+ normalised tables; query and index optimisation cut dashboard load time 60%.
- **SaaS e-commerce platform** (4-person team): Django backend and admin platform covering authentication, product, order and customer workflows plus reporting endpoints, from system design through cloud deployment.

Data Analytics Intern

Sep 2023 – Oct 2023

Vakıf Katılım Bank, Istanbul

- Supported compliance and regulatory reporting, analysing financial and operational datasets in SQL and Python for senior management and regulator-facing submissions.
- Developed 4 structured reports and dashboards monitoring regulatory indicators and operational performance.

Projects

TELCO CRM Platform

Java microservices – capstone, Turkcell GYGY 5.0 · GitHub

- Telecommunications CRM covering customer/KYC, catalog, ordering, subscriptions, usage, billing, payment, notifications and ticketing.
- Spring Cloud microservices with API Gateway, PostgreSQL database-per-service, Kafka domain events, Redis caching, JWT/OAuth2, Docker/Kubernetes, Prometheus/Grafana/OpenTelemetry.

Video Transcription & Translation Pipeline

Production tool, academic client · Python

- Python automation pipeline (FFmpeg, OpenAI Whisper, DeepL API) transcribing and translating video into multiple languages; configurable CLI, deployed in production.

Education

B.Sc. Mathematics

Graduated Feb 2026

Galatasaray University, Istanbul

- **Degree taught entirely in French** (francophone curriculum).
- **Thesis:** *Apprentissage par Renforcement* (Reinforcement Learning) – mathematical foundations (MDPs, Bellman equations, convergence), with classical RL algorithms implemented and compared on benchmark environments. Written and defended in French under Assoc. Prof. Dr. Ayşegül Ulus. Grade: AA (4.0/4.0).
- **Coursework:** Probability Theory, Linear Algebra, Numerical Analysis, Applied Mathematics, Machine Learning, Natural Language Processing, Cryptography
- **Leadership:** Mathematics Club Board Member (2021–2022), Photography Club Board Member (2018–2023); organized 35+ academic and cultural events

Training & Certifications

Turkcell GYGY 5.0, Java Development Track – 120 hours,

Mar 2026 - Jul 2026

Selected among 35 of ~5,000 applicants · Java, Spring Boot, REST APIs, PostgreSQL, Redis, Kafka, Docker, microservice architecture

IBM Data Science Professional Certificate – Coursera,

Aug 2023

Credential TK3PHHTVV248

Languages

Turkish Native

French B2+ (language of instruction, B.Sc.; thesis defended in French)

English B2+